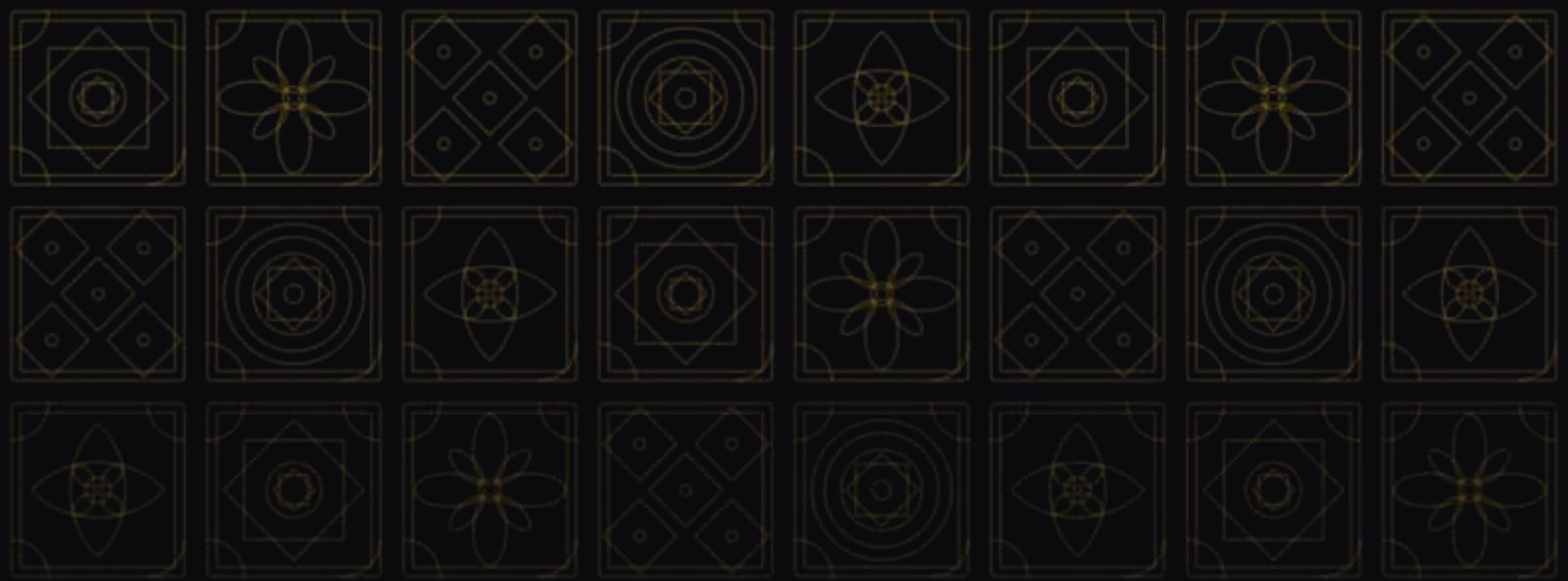




TRAINING PROGRAMME

RHCSA Part 2: Red Hat System Administration II

Secure, store, schedule and recover a Red Hat Enterprise Linux 10 server

42 HOURS

FOR Administrators who completed RHCSA Part 1 or have equivalent RHEL experience

PREPARES FOR RHCSA (EX200, RHEL 10), second half of the objectives



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Overview

This programme is the second half of the RHCSA path on Red Hat Enterprise Linux 10. Participants script administrative tasks, schedule them with cron and systemd timers, tune the system, keep SELinux enforcing, manage partitions, logical volumes and network storage, fix permission problems with ACLs, repair a system that does not boot and protect it with firewalld. They also run rootless containers with Podman, a RHEL skill that is no longer part of the RHCSA 10 exam. A timed mock EX200 exam prepares the certification, and the programme closes with a final project: a production-ready host that must survive a broken boot.

At a glance

#	PHASE	FOCUS	HOURS
01	Scripting & System Tuning	Admin scripts, regular expressions, scheduling and tuning	6 h
02	SELinux & Storage	SELinux contexts and booleans, partitions, filesystems and LVM	9 h
03	Access Control, Network Storage & Boot	ACLs, NFS and autofs, the boot process and recovery	6 h
04	Firewall, Containers & Mock Exam	firewalld, Podman containers as services, mock EX200 exam	9 h
05	Final Project	A production-ready RHEL 10 host that survives a broken boot	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
	Total		42 h

Key facts

AUDIENCE	Administrators who completed RHCSA Part 1 or have equivalent RHEL experience
PREREQUISITES	First half of the RHCSA objectives (course: RHCSA Part 1: Red Hat System Administration I) or equivalent RHEL experience with users, permissions, services, SSH and networking. These topics are not taught again
DURATION	42 h: 36 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs on RHEL 10; one timed mock exam
LAB ENVIRONMENT	Runs locally: one laptop per participant (16 GB RAM, hardware virtualization enabled) with two RHEL 10 virtual machines with spare virtual disks, built from the trainer's images and registered with each participant's no-cost Red Hat Developer Subscription for Individuals; no cloud account or cost
ASSESSMENT	A lab validated at the end of each phase, a timed mock EX200 exam and a final project graded against a checklist; certificate of completion for participants who complete the labs and the project
OUTCOME	Participants can automate, secure and troubleshoot a RHEL 10 server with SELinux enforcing, manage its storage and boot process, and sit the EX200 exam

PHASE 01

6 H

Scripting & System Tuning

Participants automate routine work with Bash scripts and regular expressions, schedule it with at, cron and systemd timers, and tune the system with tuned profiles and process priorities. Validated lab: a reporting script run every night by a systemd timer, with its output checked in the journal.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Scripting & Regex for Admins	Bash scripts, positional parameters, loops, conditionals, grep and regular expressions	Lecture + Lab	3 h
1.2	Scheduling	at, cron, systemd timer units, temporary files	Lecture + Lab	1.5 h
1.3	Tuning	tuned profiles, process scheduling, nice and renice	Lecture + Lab	1.5 h

PHASE 02

9 H

SELinux & Storage

Participants keep SELinux enforcing and fix denials instead of disabling it, then manage storage from raw disks: partitions, filesystems, swap and persistent mounts, and logical volumes that grow without downtime. Validated lab: a web server serving content from a custom directory with SELinux enforcing, on a logical volume extended while mounted.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	SELinux	Enforcing and permissive modes, file and process contexts, restorecon and semanage, booleans, troubleshooting	Lecture + Lab	3 h
2.2	Basic Storage	GPT and MBR partitions, VFAT, ext4 and XFS filesystems, swap, persistent mounts by UUID or label	Lecture + Lab	3 h
2.3	Logical Volumes	LVM physical volumes, volume groups and logical volumes, creating and extending logical volumes without data loss	Lecture + Lab	3 h

PHASE 03

6 H

Access Control, Network Storage & Boot

Participants diagnose permission problems and solve them with ACLs, mount shared storage over NFS on demand with autofs, then follow the boot process from firmware to target and recover a system that does not start. Validated lab: home directories automounted from an NFS server, and a lost root password reset on a machine that boots again cleanly.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Permission Problems & ACLs	Diagnosing and correcting permission problems, default permissions, getfacl and setfacl	Lecture + Lab	1.5 h
3.2	Network Storage	NFS mounts, autofs	Lecture + Lab	1.5 h
3.3	Boot Process	systemd targets, bootloader configuration (GRUB 2, grubby), root password reset, filesystem repair	Lecture + Lab	3 h

PHASE 04

9 H

Firewall, Containers & Mock Exam

Participants open only the services a server needs with firewalld and SELinux port labels, run rootless containers with Podman as systemd services, then sit a timed mock EX200 exam. Validated lab: a web service on a non-standard port, reachable through the firewall with SELinux enforcing, and a containerised service that starts at boot.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Network Security	firewalld zones and services, firewall-cmd, SELinux port labelling	Lecture + Lab	3 h
4.2	Containers with Podman	Images, registries, rootless containers, volumes, containers as systemd services (a RHEL skill, no longer in the RHCSA 10 exam)	Lecture + Lab	3 h
4.3	Mock Exam	Timed mock EX200 exam on RHEL 10 and debrief	Lab	3 h

PHASE 05

6 H

Final Project

Each participant turns a bare RHEL 10 virtual machine into a production-ready host from a written brief, reusing the phase labs, then proves that it can be recovered when the trainer breaks its boot.

#	CHAPTER	CONTENT	FORMAT	HOURS
5.1	Brief & Build	Read the brief, then build LVM storage and swap, the NFS automount, the SELinux-enforcing web service and the firewall rules	Project	3 h
5.2	Automate & Document	A systemd timer for maintenance, a Podman container run as a systemd service, a short runbook	Project	1.5 h
5.3	Recovery & Review	Recover from a boot failure set by the trainer, grading against the checklist, questions and feedback	Project	1.5 h

FINAL PROJECT**A Production-Ready RHEL 10 Host**

A RHEL 10 host with LVM storage and swap, an NFS automount, a web service running with SELinux enforcing behind firewalld, a maintenance job on a systemd timer and a Podman container started at boot, recovered by its builder after a broken boot.

STACK RHEL 10, Bash, systemd timers, SELinux, LVM, XFS, NFS, autofs, firewalld, GRUB 2, Podman

DELIVERABLE Configured host, runbook, graded checklist including the recovery